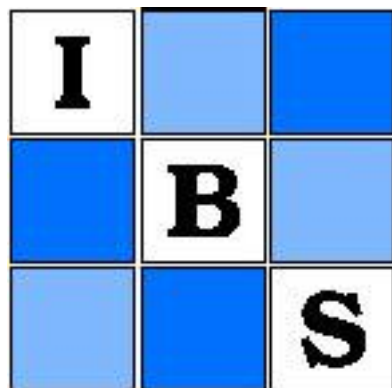




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Power/Sample Size Calculations for Generalized Linear Models

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SUMMARY

An approach for estimating power/sample size is described within the framework of generalized linear models. This approach is based on an asymptotic approximation to the power of the score test under contiguous alternatives and is applicable to tests of composite null hypotheses. An implementation is described for the special case of logistic regression models. Simulation studies are presented which indicate that the asymptotic approximation to the finite-sample situation is good over a range of parameter configurations.

1. Introduction

Many commonly used test statistics can be derived as score statistics from within the framework of generalized linear models (Gart and Tarone, 1983). In this framework independent scalar response variables, $Y_1, \dots, Y_n$, are assumed to have probability density functions of the form

$$\exp[Y_i\theta_i - b(\theta_i) + c(Y_i)]. \quad (1.1)$$

The canonical parameter θ_i is related to the expected value of Y_i , μ_i , by the equation $\mu_i = b'(\theta_i)$, where b' denotes the first derivative of b . The μ_i are in turn related to linear predictors, η_i , by the link function g in the expression $\eta_i = g(\mu_i)$. In this work, the η_i will be written as

$$\mathbf{Z}_i^T \boldsymbol{\psi} + \mathbf{X}_i^T \boldsymbol{\lambda}, \quad (1.2)$$

where $\mathbf{Z}_i$ is a p -vector of covariates, $\mathbf{X}_i$ is a q -vector of covariates, and $\boldsymbol{\psi}$ and $\boldsymbol{\lambda}$ represent the associated unknown regression coefficients. For further details about generalized linear models, see McCullagh and Nelder (1983).

We are concerned with testing the hypothesis $\boldsymbol{\psi} = \boldsymbol{\psi}_0$ while treating $\boldsymbol{\lambda}$ as a nuisance parameter. The score test of this hypothesis is based on the statistic $S_{n\boldsymbol{\psi}}(\boldsymbol{\psi}_0, \hat{\boldsymbol{\lambda}}_0)$, where $\hat{\boldsymbol{\lambda}}_0$ is a solution to the equation $S_{n\boldsymbol{\lambda}}(\boldsymbol{\psi}_0, \boldsymbol{\lambda}) = 0$ and $S_{n\boldsymbol{\psi}}$ and $S_{n\boldsymbol{\lambda}}$ represent derivatives of the log-likelihood function with respect to $\boldsymbol{\psi}$ and $\boldsymbol{\lambda}$, respectively. The actual test statistic is a quadratic form in $S_{n\boldsymbol{\psi}}(\boldsymbol{\psi}_0, \hat{\boldsymbol{\lambda}}_0)$ and is referred to its asymptotic distribution under the null hypothesis, which is a central chi-square distribution on p degrees of freedom.

In Section 2, we present an approximation to the non-null distribution of the score test statistic and describe an approach for performing sample size/power calculations. In

Key words: Generalized linear models; Noncentral chi-square; Power; Score tests.

Section 3, logistic regression is used as a specific example of how the approach can be implemented. A brief discussion of how this method may be adapted for use in unstratified analysis of case-control studies is given in Section 4. In Section 5, results from a small simulation study are presented which evaluate the adequacy of the proposed approximation for tests within logistic regression models. An example is described in Section 6.

2. An Approximation to the Power of the Score Test

The score test statistic, T_n , is computed as the quadratic form

$$\mathbf{S}_{n\psi}^T(\psi_0, \hat{\lambda}_0) \mathbf{V}_n^{-1} \mathbf{S}_{n\psi}(\psi_0, \hat{\lambda}_0), \quad (2.1)$$

where $\mathbf{V}_n$ represents an estimate of the covariance matrix of $\mathbf{S}_{n\psi}(\psi_0, \hat{\lambda}_0)$. Typically $\mathbf{V}_n$ is taken to be the appropriate submatrix of the inverse of either the observed or expected information matrix evaluated at $(\psi_0, \hat{\lambda}_0)$.

In order to approximate the distribution of T_n under alternative hypotheses, we begin by describing the limiting distribution of $\mathbf{S}_{n\psi}(\psi_0, \hat{\lambda}_0)$. Note that $\hat{\lambda}_0$ is generally not a consistent estimator of λ . It will converge to some value λ_0^* which is defined as the solution to the equation

$$\mathbf{0} = \lim n^{-1} \mathbf{E}[\mathbf{S}_{n\lambda}(\psi_0, \lambda)]. \quad (2.2)$$

Standard Taylor series arguments (Cox and Hinkley, 1974, pp. 322–324) may be used to obtain an approximation to $\mathbf{S}_{n\psi}(\psi_0, \hat{\lambda}_0)$ for which the error is $O_p(n^{-1})$. This approximation is given by

$$\mathbf{S}_{n\psi}(\psi_0, \lambda_0^*) - \mathbf{I}_{n\psi\lambda}^* \mathbf{I}_{n\lambda\lambda}^{*-1} \mathbf{S}_{n\lambda}(\psi_0, \lambda_0^*), \quad (2.3)$$

where $\mathbf{I}_{n\psi\lambda}^*$ and $\mathbf{I}_{n\lambda\lambda}^*$ represent elements of the expected information matrix evaluated at ψ_0 and λ_0^* . Let ξ_n and Σ_n denote the mean and covariance matrix of (2.3), respectively. Under mild conditions on the asymptotic stability and regularity of Σ_n , it follows that $n^{-1/2}[\mathbf{S}_{n\psi}(\psi_0, \hat{\lambda}_0) - \xi_n]$ has a limiting multivariate normal distribution with mean zero and covariance matrix given by $\lim n^{-1} \Sigma_n$.

Based on the limiting normality of $\mathbf{S}_{n\psi}(\psi_0, \hat{\lambda}_0)$ and the assumption that $\mathbf{V}_n$ provides a reasonable approximation to Σ_n , we will approximate the distribution of T_n by a chi-square distribution on p degrees of freedom with noncentrality parameter, γ_n , given by

$$\xi_n^T \Sigma_n^{-1} \xi_n. \quad (2.4)$$

A justification for this approximation is the fact that a sufficient condition for quadratic forms of multivariate normal random variables to have a chi-square distribution is that the matrix in the quadratic form be a g -inverse of the covariance matrix of the normal random variables. We propose approximating the power of the test based on T_n by evaluating γ_n for given values of ψ and λ and then referring to tables of the noncentral chi-square distribution.

There are two potential problems with this approximation. First, since $\hat{\lambda}_0$ is not a consistent estimator of λ under alternative hypotheses, $\mathbf{V}_n$ will not be a consistent estimator of Σ_n . Thus, the requirement for the distributional result cited above does not obtain, even asymptotically. Second, even though the distribution of $\mathbf{S}_{n\psi}(\psi_0, \hat{\lambda}_0)$ approaches multivariate normality, its expected value is simultaneously going off to infinity. Therefore, the result relies on the quality of the chi-square approximation at a sequence of points that move progressively farther out in the tail of the distribution as n becomes large.

These potential problems are not likely to be serious for alternatives close to the null hypothesis since the noncentral chi-square distribution is asymptotically correct under sequences of contiguous alternatives (Cox and Hinkley, 1974). However, most sample size/power calculations are performed at alternatives that are not close to the null

hypothesis. It is unclear whether the approximation described above will be good for these alternatives.

In the special case of generalized linear models, the form of $\mathbf{S}_{n\psi}$ and $\mathbf{S}_{n\lambda}$ implied by (1.1) may be used to write expression (2.3) as

$$\sum_{i=1}^n [Y_i - \mu_i(\psi_0, \lambda_0^*)] \Delta_i \mathbf{Z}_i^* \tag{2.5}$$

where Δ_i is the first derivative of θ_i with respect to η_i , $\mathbf{Z}_i^*$ represents $\mathbf{Z}_i - \mathbf{I}_{n\psi\lambda}^* \mathbf{I}_{n\lambda\lambda}^{*-1} \mathbf{X}_i$, and μ_i is written to show explicitly its dependence on ψ and λ . It then follows that the expected value, ξ_n , and covariance matrix, Σ_n , of (2.3) are given by the expressions

$$\sum_{i=1}^n [\mu_i(\psi, \lambda) - \mu_i(\psi_0, \lambda_0^*)] \Delta_i \mathbf{Z}_i^* \tag{2.6}$$

and

$$\sum_{i=1}^n v_i(\psi, \lambda) \Delta_i^2 \mathbf{Z}_i^* \mathbf{Z}_i^{*T} \tag{2.7}$$

where $v_i(\psi, \lambda) = \text{var}(y_i)$.

3. An Implementation

We will make the simplifying assumption that all of the covariates in the model are categorical. This implies that there are a finite number of distinct covariate configurations $\{(\mathbf{Z}_i, \mathbf{X}_i); i = 1, \dots, m\}$ with sample proportions at each configuration $\{\Pi_i; i = 1, \dots, m\}$.

The first step in the calculations is to specify values for the parameters ψ , λ , and $\{\Pi_i; i = 1, \dots, m\}$. Often the regression parameters associated with covariates are straightforward to specify but the intercept parameter is not. If the average response is specified in addition to the regression parameters, then the intercept parameter in the model can be found as the solution to the equation

$$\bar{\mu} = \sum_{i=1}^m \Pi_i \mu_i(\psi, \lambda), \tag{3.1}$$

where $\bar{\mu}$ denotes the population mean response.

Once parameter values for ψ and λ are chosen, λ_0^* can be computed as the solution to the equation

$$0 = \sum_{i=1}^m \Pi_i [\mu_i(\psi, \lambda) - \mu_i(\psi_0, \lambda_0^*)] \Delta_i \mathbf{X}_i. \tag{3.2}$$

Solving equation (3.2) is equivalent to fitting the null model in a weighted analysis where the data are taken to be $\{\mu_i(\psi, \lambda); i = 1, \dots, m\}$ and the weights are $\{\Pi_i; i = 1, \dots, m\}$. This calculation can typically be performed with the same computer software that is used to perform the data analysis. Note that λ_0^* does not depend on n .

Rewriting expressions (2.6) and (2.7) as sums over covariate configurations rather than sums over individuals, we have $\xi_n = n\xi_0$ and $\Sigma_n = n\Sigma_0$, where

$$\xi_0 = \sum_{i=1}^m \Pi_i [\mu_i(\psi, \lambda) - \mu_i(\psi_0, \lambda_0^*)] \Delta_i \mathbf{Z}_i^* \tag{3.3}$$

and

$$\Sigma_0 = \sum_{i=1}^m \Pi_i v_i(\psi, \lambda) \Delta_i^2 \mathbf{Z}_i^* \mathbf{Z}_i^{*T}. \tag{3.4}$$

Then $\gamma_n = n\gamma_0$ where γ_0 is given by

$$\xi_0^T \Sigma_0^{-1} \xi_0. \quad (3.5)$$

In the special case of logistic regression, the response variable, Y_i , follows a Bernoulli distribution with probability of response μ_i . The link function is the logistic function $\log(\mu_i) - \log(1 - \mu_i)$. In order to compute the intercept parameter given values for the other parameters and a value for the overall probability of response, equation (3.1) can be easily solved with the iteration scheme

$$\alpha^{(k)} = \log(\bar{\mu}) - \log \left\{ \sum_{i=1}^m \Pi_i \exp(\mathbf{Z}_i^T \psi + \mathbf{X}_i^T \lambda) / [1 + \exp(\alpha^{(k-1)} + \mathbf{Z}_i^T \psi + \mathbf{X}_i^T \lambda)] \right\} \quad (3.6)$$

where $\alpha^{(k)}$ denotes the estimate of the intercept at the k th iteration and, in this expression, $\mathbf{X}_i$ and λ do not include the coordinate associated with the intercept. The iterations may be started with the initial value, $\alpha^{(0)}$, given by

$$\log(\bar{\mu}) - \log(1 - \bar{\mu}) - \sum_{i=1}^m \Pi_i (\mathbf{Z}_i^T \psi + \mathbf{X}_i^T \lambda).$$

4. Unmatched Case-Control Studies

Logistic regression models are often used to analyze data from unmatched case-control studies. Because the prospective logistic regression model may be used to analyze these data in spite of the retrospective nature of the sampling (Prentice and Pyke, 1979), the approximation described above could be used to estimate the power of these studies. The main difficulty in adapting the method to be used in case-control studies is in specification of the joint distribution of $\mathbf{Z}$ and $\mathbf{X}$.

One approach to this problem is to specify the distribution of $(\mathbf{Z}, \mathbf{X})$ over the sample of cases. Then the specified parameters in the prospective model relating $\mathbf{Z}$ and $\mathbf{X}$ to the probability of response are used to compute the distribution of $(\mathbf{Z}, \mathbf{X})$ over the sample of controls. The Π_i 's could then be calculated as a mixture of the $(\mathbf{Z}, \mathbf{X})$ distributions over the case and control samples.

To be specific, let $\{\Pi_{1i}: 1 \leq i \leq m\}$ denote the distribution of $(\mathbf{Z}, \mathbf{X})$ over cases in the population. A simple calculation indicates that the distribution of $(\mathbf{Z}, \mathbf{X})$ among noncases in the population, denoted Π_{2i} , is proportional to $\{\Pi_{1i} \exp(-\eta_i): 1 \leq i \leq m\}$, where η_i denotes the predictor for the i th covariate configuration but excluding the intercept term. The distribution of $(\mathbf{Z}, \mathbf{X})$ over the sample that is used in the above method may then be calculated as a weighted average of Π_{1i} and Π_{2i} with the weights being the fraction of cases and noncases in the sample.

5. A Simulation Study

In order to evaluate the adequacy of the approximation, we used the method described above to calculate sample sizes required to achieve selected size and power characteristics for tests within logistic regression models. We then performed computer simulations to estimate the true power that is obtained when the calculated sample sizes are used. Each Monte Carlo estimate of power was based on 3,000 independently generated data sets. All computer simulations were performed with FORTRAN programs running on an IBM PC/AT using the random number generator described by Wichmann and Hill (1982).

One advantage of the proposed method for determining sample sizes is that the effect of adjustment for confounding variables can be incorporated into the calculations. In order to evaluate the proposed method in this situation, we consider tests of the same size and

nominal power as above but with an exposure variable, Z , and a single confounder variable, X , each taking values of either zero or one. Their joint distribution over the sample is specified by Π_1, Π_2, Π_3 , and Π_4 , which correspond to (Z, X) values of $(0, 0), (0, 1), (1, 0)$, and $(1, 1)$, respectively. The odds ratio associated with the confounder X , $\exp(\lambda)$, is taken to be 2.0. Odds ratios associated with the exposure variable Z , $\exp(\psi)$, are taken to be 2.0 and 3.0. The average response probability, $\bar{\mu}$, is taken to be .50 and .15.

Tables 2 and 3 present sample sizes computed by the proposed method and Monte Carlo estimates of power for tests of size .05 and .01, respectively. Over the range of conditions

Table 1
Calculated sample sizes and Monte Carlo estimates of power for the two-sample problem

$\bar{\mu}$	e^ψ	Nominal power	$\alpha = .05$		$\alpha = .01$	
			N	Power (SE)	N	Power (SE)
.15	2.0	.80	531	79.6 (.74)	789	81.1 (.71)
		.90	710	90.6 (.53)	1,005	89.4 (.56)
		.95	878	95.7 (.37)	1,203	95.6 (.38)
		.99	1,242	99.1 (.18)	1,624	98.7 (.21)
	3.0	.80	223	81.0 (.72)	331	80.3 (.73)
		.90	298	90.2 (.54)	422	90.7 (.53)
		.95	368	95.1 (.39)	505	95.5 (.38)
		.99	521	99.1 (.18)	681	99.6 (.11)
	2.0	.80	259	80.5 (.72)	385	78.6 (.75)
		.90	347	89.3 (.56)	491	89.5 (.56)
		.95	429	93.9 (.44)	588	94.9 (.40)
		.99	606	98.8 (.20)	793	98.5 (.22)
.50	3.0	.80	102	80.7 (.72)	151	80.4 (.72)
		.90	136	87.5 (.60)	193	89.2 (.57)
		.95	168	94.1 (.43)	231	94.0 (.43)
		.99	238	98.7 (.21)	311	99.2 (.17)

Table 2
Calculated sample sizes and Monte Carlo estimates of power for score test with size .05

$\bar{\mu}$	e^ψ	Nominal power	$(\Pi_1, \Pi_2, \Pi_3, \Pi_4)$					
			$(.25, .25, .25, .25)$		$(.4, .1, .1, .4)$		$(.1, .4, .4, .1)$	
			N	Power (SE)	N	Power (SE)	N	Power (SE)
.15	2.0	.80	533	79.3 (.74)	842	78.8 (.75)	882	79.9 (.73)
		.90	714	89.5 (.56)	1,128	88.3 (.59)	1,100	91.0 (.52)
		.95	882	95.4 (.38)	1,395	94.3 (.42)	1,361	94.6 (.41)
		.99	1,247	98.9 (.19)	1,972	98.9 (.19)	1,924	99.0 (.18)
	3.0	.80	221	80.6 (.72)	356	79.2 (.74)	339	82.1 (.70)
		.90	296	89.7 (.56)	477	88.8 (.58)	453	90.9 (.53)
		.95	366	95.0 (.40)	590	94.9 (.40)	561	95.9 (.36)
		.99	517	99.0 (.18)	834	98.8 (.20)	793	99.4 (.14)
	2.0	.80	267	80.3 (.73)	432	81.4 (.71)	403	77.8 (.76)
		.90	358	89.8 (.55)	578	91.3 (.52)	539	89.0 (.57)
		.95	442	94.8 (.41)	715	95.3 (.39)	666	93.9 (.44)
		.99	625	98.7 (.20)	1,011	99.0 (.18)	942	98.7 (.20)
.50	3.0	.80	105	80.1 (.73)	173	82.3 (.70)	155	77.0 (.77)
		.90	141	89.4 (.56)	232	91.0 (.52)	208	87.9 (.60)
		.95	174	94.9 (.40)	286	96.0 (.36)	257	93.6 (.45)
		.99	246	98.7 (.21)	405	99.2 (.16)	363	98.2 (.24)

Table 3
Calculated sample sizes and Monte Carlo estimates of power for score test with size .01

$\bar{\mu}$	e^{ψ}	Nominal power	(II ₁ , II ₂ , II ₃ , II ₄)					
			(.25, .25, .25, .25)		(.4, .1, .1, .4)		(.1, .4, .4, .1)	
			N	Power (SE)	N	Power (SE)	N	Power (SE)
.15	2.0	.80	793	79.0 (.74)	1,253	77.8 (.76)	1,223	81.7 (.71)
		.90	1,010	90.1 (.55)	1,596	89.1 (.57)	1,558	90.7 (.53)
		.95	1,209	95.3 (.39)	1,911	94.5 (.42)	1,865	95.7 (.37)
		.99	1,631	99.1 (.17)	2,578	98.6 (.21)	2,516	99.2 (.16)
	3.0	.80	328	79.9 (.73)	530	76.3 (.78)	504	80.9 (.72)
		.90	418	89.3 (.56)	675	88.3 (.59)	642	91.1 (.52)
		.95	501	94.7 (.41)	808	94.7 (.41)	768	95.9 (.36)
		.99	676	99.3 (.15)	1,090	99.0 (.18)	1,036	98.8 (.20)
	.50	.80	397	79.0 (.74)	643	81.6 (.71)	599	76.9 (.77)
		.90	506	89.4 (.56)	818	90.5 (.53)	763	87.7 (.60)
		.95	606	94.7 (.41)	980	95.8 (.37)	913	93.6 (.45)
		.99	817	99.0 (.18)	1,322	99.1 (.18)	1,232	98.4 (.23)
	3.0	.80	156	78.4 (.75)	257	83.7 (.67)	231	75.4 (.79)
		.90	199	90.0 (.55)	328	91.9 (.50)	294	85.0 (.65)
		.95	238	94.2 (.43)	392	95.6 (.37)	351	92.6 (.48)
		.99	321	98.7 (.21)	529	99.3 (.15)	474	98.0 (.25)

considered here, the approximation described above performs very well. The worst approximation occurs in the situation where there is a negative correlation between the exposure and confounder variable and the average probability of response is .50. In this situation, the estimated actual power is as much as .05 below the nominal power.

In comparing Table 1 to Tables 2 and 3, we see that the sample sizes required to achieve a specified power generally increase when estimation of the effects of confounding variables is required. When there is no correlation between the exposure and confounder variables, the sample sizes are virtually the same as in the case without a confounder. As expected, when there is correlation between the exposure and confounding variables, the sample sizes increase dramatically.

The quality of the approximation for retrospective sampling schemes was evaluated by simulation under several configurations and was found to be comparable to the results for prospective sampling described above.

6. An Example

A large randomized clinical trial funded by the National Cancer Institute to investigate the relationship between dietary fat and risk of breast cancer in women at an elevated risk for breast cancer is currently underway. Approximately 1,000 cases of breast cancer are expected to be detected over the duration of the trial. At the conclusion of the trial, it is anticipated that a variety of secondary analyses will be performed relating suspected etiologic risk factors to breast cancer incidence after taking into account the effects of known risk factors for breast cancer. It would be useful to obtain estimates of power for these tests.

For illustrative purposes, consider a test for the association between use of oral contraceptives and breast cancer after taking into account the effect of the known risk factor age at first live birth. We will assume that the test will be performed as part of an unmatched case-control analysis with one control per case. An estimate of the joint distribution of oral contraceptive use and age at first live birth, which was derived from preliminary data from

Table 4
Joint distribution of oral contraceptive use and age at first live birth

Age at first live birth	Oral contraceptive use		Odds ratio associated with age at first live birth
	No	Yes	
<20	.11	.07	1.00
20–24	.28	.17	1.31
25–29	.13	.08	1.71
≥30	.05	.03	2.23
Nulliparous	.06	.02	1.71

the aforementioned trial, is given in Table 4. Estimates of the odds ratios associated with each category of age at first live birth are also given in Table 4. These estimates are taken from Bruzzi et al. (1985). Using these parameters, we estimate that the odds ratios associated with oral contraceptive use that can be detected by a test of size .05 with power .80, .90, and .95 are 1.29, 1.34, and 1.39, respectively. Taking into account the adjustment for age at first live birth in this test increases the magnitude of the odds ratios that are detectable with the specified power to 1.36, 1.42, and 1.48, respectively.

7. Discussion

The method for approximating power of score tests that is described here is very general and quite flexible. In addition, the accuracy of the approximation appears to be excellent for logistic regression models over the range of conditions that we have considered. Implementation of the method for tests based on other generalized linear models such as log-linear models with Poisson errors is straightforward. Verification of the accuracy of the approximation under a variety of different models would be useful.

In performing the simulation studies, we estimated the power of the likelihood-ratio test in order to compare it to the power of the score test. It is expected that the powers of the two tests would be quite similar at alternatives close to the null hypothesis due to the local optimality properties of the score test. However, at more distant alternatives, we expected the likelihood-ratio test to enjoy greater power. We found that the powers of the two tests were virtually identical under all of the conditions studied.

RÉSUMÉ

On décrit une approche pour estimer la puissance et la taille d'un échantillon, dans le domaine des modèles linéaires généralisés. Cette approche est basée sur une approximation asymptotique de la puissance du test des scores, sous des alternatives proches, et est applicable pour des tests d'hypothèses nulles composées. On décrit une réalisation, dans le cas particulier des modèles de régression logistique. On présente des études de simulation, et on montre que l'approximation asymptotique, pour un échantillon de taille finie, est bonne pour un ensemble de situations.

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APPENDIX

The procedure for estimating power may be performed by the following five steps:

1. Specify parameter values (ψ, λ) and the anticipated sample proportions $\{\Pi_i; i = 1, \dots, m\}$ at each of the covariate configurations $\{(Z_i, X_i); i = 1, \dots, m\}$.
2. Compute the expected value of response at each covariate configuration, $\{\mu_i; i = 1, \dots, m\}$ from (ψ, λ) and $\{(Z_i, X_i); i = 1, \dots, m\}$.
3. Form a data set with m "observations" in which the i th observation has response μ_i , covariate values (Z_i, X_i) and prior weight Π_i . Fit the null model to this data to obtain an estimate of λ, λ_0^* . Compute the expected value of response at each covariate configuration, $\{\mu_i^*; i = 1, \dots, m\}$ from the parameter value (ψ, λ_0^*) and $\{(Z_i, X_i); i = 1, \dots, m\}$.
4. Calculate the score vector for ψ from this data set under the model with parameter values (ψ, λ_0^*) , $\sum \Pi_i(\mu_i - \mu_i^*)\Delta_i Z_i$, and compute the quadratic form in this vector with matrix given by the inverse of

$$\begin{pmatrix} \mathbf{I} & \mathbf{I}_{\psi\lambda}^* \mathbf{I}_{\lambda\lambda}^{*-1} \end{pmatrix} \begin{pmatrix} \mathbf{I}_{\psi\psi} & \mathbf{I}_{\psi\lambda} \\ \mathbf{I}_{\lambda\psi} & \mathbf{I}_{\lambda\lambda} \end{pmatrix} \begin{pmatrix} \mathbf{I} \\ \mathbf{I}_{\lambda\lambda}^{*-1} \end{pmatrix} \begin{pmatrix} \mathbf{I}_{\lambda\psi}^* \\ \mathbf{I}_{\lambda\psi}^{*T} \end{pmatrix},$$

where $\mathbf{I}$ is the identity matrix and $\mathbf{I}, \mathbf{I}^*$ represent elements of the expected information matrix evaluated at (ψ, λ) and (ψ, λ_0^*) , respectively. To be specific, $\mathbf{I} = \sum \Pi_i v_i(\psi, \lambda) \Delta_i^2 (Z_i/X_i)^{\otimes 2}$ and $\mathbf{I}^* = \sum \Pi_i v_i(\psi, \lambda_0^*) \Delta_i^2 (Z_i/X_i)^{\otimes 2}$, where $v_i(\psi, \lambda)$ denotes the variance of the i th response if the true parameter values are ψ and λ .

5. Enter a table of tail probabilities for a chi-square distribution with p degrees of freedom and noncentrality parameter given by n times the quadratic form computed in step 4 to obtain the estimate of power.